| Exp No: 4 | **Setup and configure a LAN using a Switch and Ethernet cables** |
|---|---|

## AIM:

To set up and configure a LAN using a switch and Ethernet cables, assign IP addresses to the hosts, check the connectivity using the ping command and share files and folders across the machines of the LAN.

## ALGORITHM / PROCEDURE:

1. Connect 4 host machines to the switch using straight-through UTP cables (RJ45). Check that the link LEDs of the switch glow.
2. Assign a static IP address to each host: **Windows**, Control Panel > Network and Sharing Centre > Change adapter settings > Ethernet > Properties > IPv4 (or run `ncpa.cpl`).
3. Use the addresses `192.168.10.11` to `192.168.10.14` with subnet mask `255.255.255.0`. Leave the default gateway empty because there is only one network.
4. Open the Command Prompt and verify the settings with `ipconfig`.
5. Check the connectivity between the machines with `ping`, for example from PC1 `ping 192.168.10.12`.
6. On PC1 create a folder `LabShare`, right click > Properties > Sharing > Advanced Sharing > tick **Share this folder** and give **Read/Write** permission to **Everyone**.
7. On PC2 open **File Explorer** and type `\\192.168.10.11` in the address bar. The files of PC1 are displayed.
8. The folder can also be mapped as a drive with `net use Z: \\192.168.10.11\LabShare` and the shared files can be copied between the machines.
9. Note down the output of each command.

| Host | IP address | Subnet mask | Switch port |
|---|---|---|---|
| PC1 | `192.168.10.11` | `255.255.255.0` | Fa0/1 |
| PC2 | `192.168.10.12` | `255.255.255.0` | Fa0/2 |
| PC3 | `192.168.10.13` | `255.255.255.0` | Fa0/3 |
| PC4 | `192.168.10.14` | `255.255.255.0` | Fa0/4 |

## DIAGRAM:

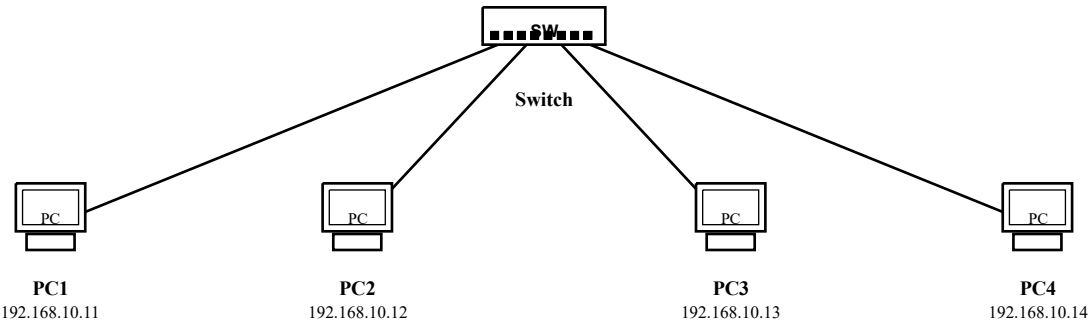



**Fig 1: LAN with 4 hosts connected to a switch (star topology)**

# OUTPUT:

**Command Prompt - PC1**

```
C:\Users\userUSER >ipconfig

Windows IP Configuration


Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::6c1f:3a2e:91b4:d7c2%8
   IPv4 Address. . . . . . . . . . . : 192.168.10.11
Subnet Mask . . . . . . . . . . . . : 255.255.255.0
Default Gateway . . . . . . . . . :
```

**Command Prompt - PC1**

```
C:\Users\userUSER >ping 192.168.10.12 Pinging

192.168.10.12 with 32 bytes of data:

Reply from 192.168.10.12: bytes=32 time=1ms TTL=128
Reply from 192.168.10.12: bytes=32 time<1ms TTL=128
Reply from 192.168.10.12: bytes=32 time<1ms TTL=128
Reply from 192.168.10.12: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.12:     Packets: Sent =
4, Received = 4, Lost = 0 (0% loss), Approximate round
trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\Users\userUSER >ping 192.168.10.13 Pinging

192.168.10.13 with 32 bytes of data:

Reply from 192.168.10.13: bytes=32 time<1ms TTL=128
Reply from 192.168.10.13: bytes=32 time<1ms TTL=128
Reply from 192.168.10.13: bytes=32 time=1ms TTL=128
Reply from 192.168.10.13: bytes=32 time<1ms TTL=128 Ping

statistics for 192.168.10.13:

    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate
round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\Users\userUSER >ping 192.168.10.14 Pinging

192.168.10.14 with 32 bytes of data:

Reply from 192.168.10.14: bytes=32 time<1ms TTL=128
Reply from 192.168.10.14: bytes=32 time<1ms TTL=128
Reply from 192.168.10.14: bytes=32 time<1ms TTL=128
Reply from 192.168.10.14: bytes=32 time<1ms TTL=128 Ping

statistics for 192.168.10.14:

    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

**Command Prompt - PC2**

```
C:\Users\userUSER >net view \\192.168.10.11
Shared resources at \\192.168.10.11

Share name   Type   Used as   Comment
-------------------------------------------------
- LabShare     Disk The command completed
successfully.

C:\Users\userUSER >net use Z: \\192.168.10.11\LabShare The
command completed successfully.

C:\Users\userUSER >dir Z:
 Volume in drive Z is LabShare
 Directory of Z:\ 07-10-2026  08:42 PM    <DIR>
.
```

**Command Prompt - PC2 (contd.)**

```
07-10-2026  08:42 PM    <DIR>          ..
07-10-2026  08:40 PM            14,336 CN_Lab_Notes.docx
07-10-2026  08:41 PM           212,480 Exp4_Topology.png
07-10-2026  08:42 PM             1,024 hosts_list.txt
               3 File(s)        227,840 bytes
               2 Dir(s)  48,213,327,872 bytes free

C:\Users\userUSER >copy Z:\hosts_list.txt C:\Users\userUSER
        \Desktop         1 file(s) copied.
```

## RESULT:

Thus a LAN was set up with 4 hosts connected to a switch using Ethernet cables, the IP addresses were assigned, the connectivity was verified using the ping command and files were shared and accessed across the machines of the LAN.